

Scalable Wafer-Level Fabrication of Slanted TiO_2 Gratings for Directional Visible Light Control

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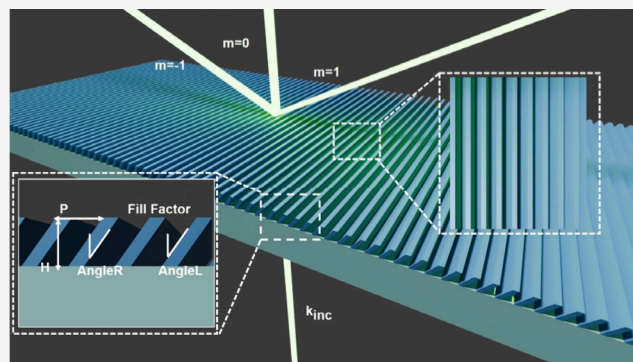
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Supporting Information

ABSTRACT: Slanted TiO_2 gratings serve as key couplers enabling high-efficiency, large-angle diffraction in augmented reality (AR) diffractive optical waveguide systems, thereby supporting compact and transparent display integration. However, fabrication challenges and high production costs have hindered their scalability. Here, we demonstrate a streamlined and high-fidelity pattern transfer framework, integrating interference lithography, nanoimprint lithography, inductively coupled plasma etching, and reactive ion beam etching. This framework enables one-step pattern replication and precise angular control across wafer-scale areas. An experimental demonstration of the slanted TiO_2 grating confirms high structural fidelity and effective phase modulation, enabling efficient directional light steering toward desired diffraction orders. Our work offers a practical and scalable fabrication strategy for implementing slanted TiO_2 gratings in AR systems, advancing their feasibility for large-scale integration and commercial deployment.

KEYWORDS: slanted TiO_2 grating, wafer-level fabrication, large-angle diffraction, diffractive optical waveguide



Augmented Reality (AR) has emerged as a revolutionary display technology with broad application prospects in entertainment, education, and human–machine interaction systems.^{1–4} Among various near-eye display configurations (such as axial beam splitters,⁵ cascaded mirror arrays,⁶ and diffractive optical waveguide (DOW)^{7,8}), DOW is regarded as the most promising solution due to its compact form factor and compatibility with conventional optical glass. Driven by micronano optics and nanofabrication techniques, the DOW has evolved into a highly practical approach for industrial deployment. In such systems, display light is coupled into a planar waveguide at a specific angle via in-coupler structures, propagates by means of total internal reflection (TIR), and is ultimately extracted by an out-coupler region to form the AR image visible to the user.^{9,10}

The coupling structures play a pivotal role in controlling the light path within the DOW, as they must redirect incident light beyond the critical angle to maintain TIR.¹¹ Consequently, various design strategies have been proposed, including metasurfaces based on generalized Snell's law, which enable electromagnetic wavefront control.^{12,13} However, such design principles typically suffer from a trade-off between diffraction efficiency and diffraction angle, particularly at large angles required for compact DOW geometries. In pursuit of higher angular performance, recent studies have investigated topological optical structures and in-plane asymmetry designs to

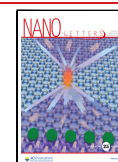
suppress undesired diffraction orders and enhance target diffraction channels.^{14–16} Despite their conceptual advantages, these strategies suffer from low scalability and complex fabrication requirements. Many rely on specially customized subwavelength unit cells, such as asymmetric configurations formed by combining two distinct nanostructures, that can only be realized with the design freedom of electron-beam lithography (EBL) or focused ion beam (FIB). However, the reliance on such serial writing techniques inevitably hinders scalable fabrication. In contrast, diffractive gratings with tailored structural asymmetry offer a fabrication-friendly and industrially scalable alternative. By breaking structural symmetry through out-of-plane asymmetric profiles, such as blazed gratings^{17,18} and slanted gratings,^{7,19} these gratings enable directional light steering while naturally suppressing unwanted diffraction orders. Among them, slanted gratings are particularly attractive for in-coupling applications due to their high efficiency (>80%). Moreover, recent studies have shown that combining in-plane and out-of-plane asymmetry can yield

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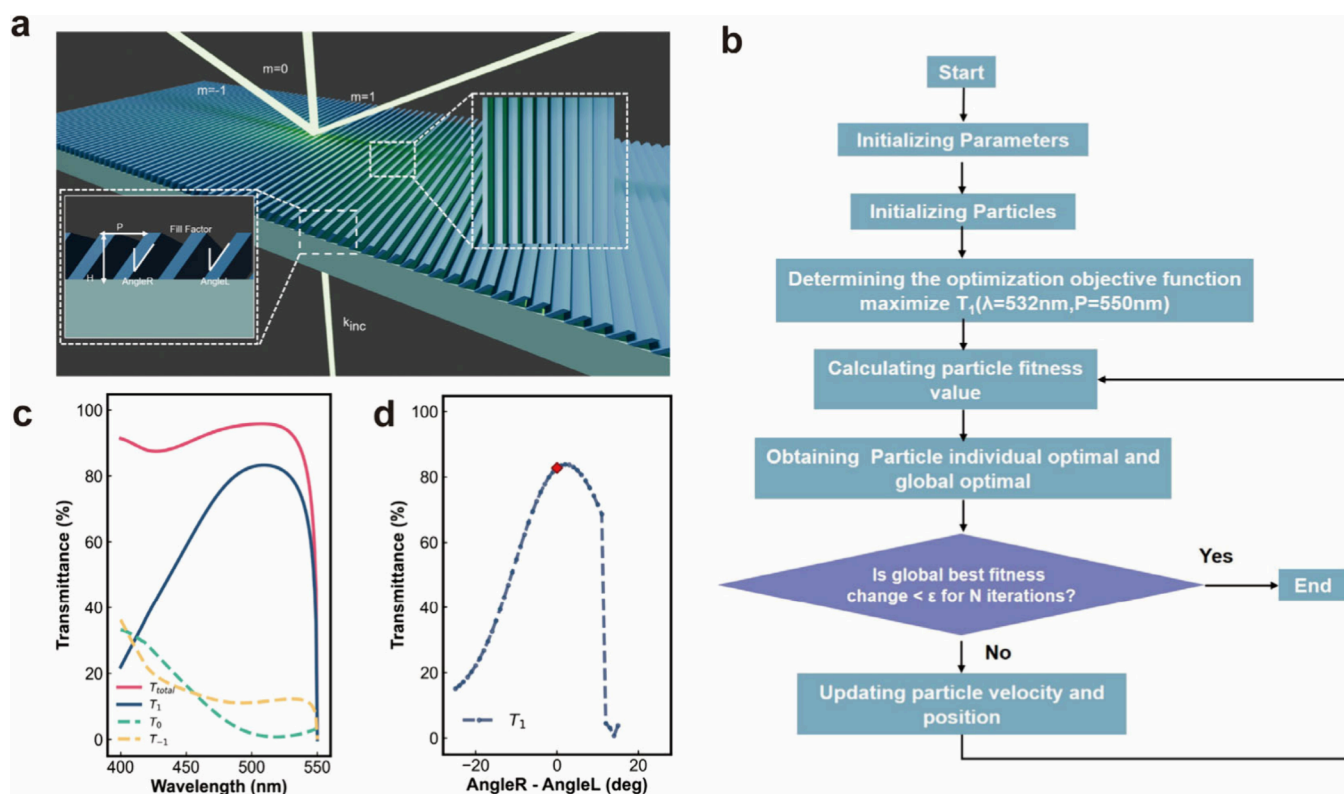


Figure 1. (a) Schematic of the slanted grating and design parameters, including period, height, fill factor and slanted angles (AngleL and AngleR). (b) Flowchart of the particle swarm optimization (PSO) used to optimize the + first order transmission efficiency (T_1). (c) Numerically calculated wavelength-dependent 0th (T_0), + first order (T_1), -first order (T_{-1}), and the total (T_{total}) transmission efficiency under TM-polarized 532 nm illumination. Calculation parameters: $p = 550$ nm; fill factor = 0.32; $H = 350$ nm; $\text{AngleL} = \text{AngleR} = 30^\circ$. (d) Impact of sidewall deviation $\text{AngleR} - \text{AngleL}$ (with AngleL fixed at 30°) on T_1 . A 10° mismatch leads to a 40% drop in T_1 diffraction efficiency.

exceptional performance with reported absolute efficiency up to 83.6% at diffraction angles as large as 80° with robust process compatibility.²⁰

However, fabricating slanted gratings in high refractive index dielectrics like TiO_2 remains technically challenging, as these subwavelength structures require high-resolution lithography techniques. Patterning methods include EBL,²¹ focused ion beam FIB,²² interference lithography (IL)²³ and nanoimprint lithography (NIL).²⁴ Among them, IL and NIL inherently support large-area fabrication, offering a scalable route beyond serial writing techniques. In addition to lithographic considerations, TiO_2 stands out as the ideal high refractive index dielectric for visible-spectrum applications due to its excellent optical properties.²⁵ Although efforts have been made to improve the refractive index of polymers or sol-gel materials through composite formulations or doping,^{26–29} few alternatives match the performance of TiO_2 in the visible regime.^{30,31} Despite these material advantages, conventional fabrication methods for producing slanted TiO_2 gratings, such as directional ICP etching, suffer from critical limitations. These include limited angle control (e.g., 10° deviation from a designed 30° slant),²⁰ structural nonuniformity, and geometric constraints imposed by etching chamber configurations,^{7,32} making them impractical for reliable wafer-level fabrication.

To address these issues, we propose a scalable wafer-level fabrication strategy for slanted TiO_2 gratings operating in the visible regime. Our approach combines IL for large-area patterning, ICP trimming for profile optimization, NIL for high-throughput pattern replication and finally RIBE for precise pattern transfer into TiO_2 . RIBE offers superior

anisotropy and angle control by using tailored gas mixtures including inert gases (e.g., Ar) and reactive gases (e.g., SF_6 , CHF_3 , CF_4).^{17,33–36} This hybrid process combination enables the fabrication of centimeter-scale slanted gratings with excellent uniformity and design flexibility. While each of these fabrication techniques is well established, their integration into a unified process tailored to overcome the etching and fidelity challenges of slanted TiO_2 structures has not, to our knowledge, been previously demonstrated. We further introduce a design methodology based on Rigorous Coupled-Wave Analysis (RCWA)³⁷ and particle swarm optimization (PSO)³⁸ to target large-angle diffraction for AR waveguide applications which is a key requirement for efficient in-coupling in an AR waveguide system. Experimentally, we fabricated a standalone slanted TiO_2 grating designed for large-angle diffraction. The fabricated gratings for large angle diffraction exhibited a first-order diffraction angle of 75° with 39.1% efficiency under 532 nm TM-polarized normal incidence. Although the grating was initially characterized as a standalone structure, we further simulated its performance when integrated into a DOW system. The results confirm that the fabricated grating design enables directional coupling and efficient in-coupling behavior under AR-relevant configurations. These combined experimental and simulation results demonstrate the feasibility and application potential of our fabrication and design strategy for large-angle, high-efficiency coupling in next-generation AR systems.

Figure 1a illustrates the schematic of the slanted TiO_2 grating designed in this work, which features a deliberate out-of-plane asymmetry introduced by tilting the grating

sidewalls at an angle. Unlike conventional vertical gratings exhibiting out-of-plane symmetry, this slanted geometry induces a spatially varying effective index along the light propagation direction. Consequently, the grating imparts a linear phase modulation, enabling directional redistribution of the optical field. The corresponding phase profile, shown in Figure S1c, exhibits a monotonic and nearly linear trend, in stark contrast to the stepped phase pattern of vertical gratings. Such continuous phase modulation favors the suppression of undesired diffraction orders and enhances large-angle diffraction efficiency.^{20,39} As shown in Figure S2, the horizontal component of the Poynting vector (x direction) exhibits a clear asymmetry along the x -axis, indicating a preferential energy flow toward the + first order diffraction direction. In contrast, the vertical component (y direction) remains nearly symmetric. This directional energy distribution reflects the influence of the slanted grating's out-of-plane asymmetry, which breaks mirror symmetry and biases diffraction toward positive orders.

We employed a well-established RCWA-PSO optimization framework (Figure 1b) to numerically determine the optimal parameter combination. The objective of the optimization was to maximize the + first order transmission efficiency (T_1). To achieve this, we defined the objective function as $\text{Min}(|1 - T_1|)$ to determine the optimal structural parameters for operation at a wavelength of 532 nm. Based on the grating equation under normal incidence, a grating period of 550 nm was selected to achieve an absolute diffraction angle of approximately 75° for both the + first and −first orders (T_0 and T_{-1}) at 532 nm (as shown in Supporting Information). The optimized structural parameters were as follows: fill factor = 0.32, grating height = 350 nm, and slant angle = 30° . The dependence of T_1 on these parameters is shown in Figure S3. At the optimized point and a wavelength of 532 nm, T_1 reaches 80% (Figure 1c), while T_0 and T_{-1} are suppressed.

Furthermore, Figure 1d demonstrates the sensitivity of performance to fabrication errors in the slant angle. Specifically, introducing a 10° deviation where $\text{AngleR} = 20^\circ$ (with AngleL fixed at 30°) results in a significant $\sim 40\%$ relative reduction in T_1 . This result highlights the critical importance of maintaining angular symmetry ($\text{AngleL} \approx \text{AngleR}$) for achieving high diffraction performance.

Building upon the previous design framework, this proposed design method can be extended to the DOW applications. Figure 2a depicts a planar DOW that integrates slanted TiO_2 gratings as in-coupler and out-coupler elements. This architecture enables light injection from a microdisplay into the user's field of view via TIR. Given the waveguide's millimeter-scale thickness and high refractive index ($n = 2.0$), geometrical ray optics can be adopted to model the beam propagation between the in-coupler and out-coupler.⁴⁰ To ensure uniform output-efficiency, the out-coupler region was divided into four segments. Without such division, nonuniform intensity distribution will be caused due to propagation attenuation as shown in Figure S4. Consequently, each segment was independently optimized using the PSO framework introduced in Figure 1b, targeting a balance between T_1 and zeroth order reflectance efficiency (R_0), as shown in Figure 2b. This segmentation strategy balances light extraction along the waveguide and expands the effective eye box. Details of the optimization method are provided in the Supporting Information.

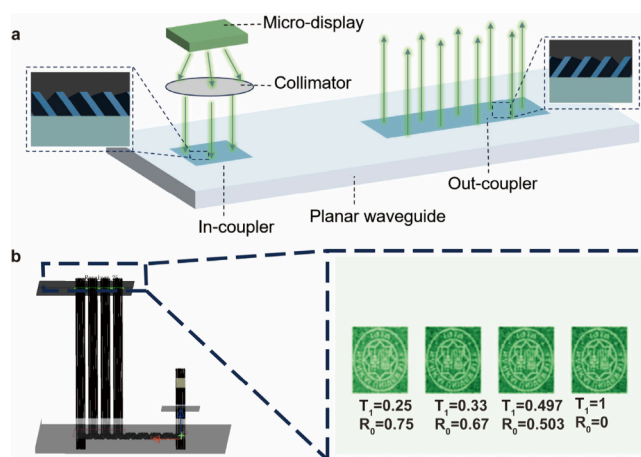


Figure 2. (a) Schematic of a one-dimensional pupil-expanded diffraction optical waveguide. (b) Numerical simulation schematic showing the out-coupler divided into four segments. Each segment was independently optimized for + first-order transmittance (T_1) and 0th-order reflectance (R_0) using the PSO framework in Figure 1b, ensuring uniform light extraction across the output field. Logo reproduced with permission of Nankai University. © 2025 Nankai University.

This formulation ensures that each grating unit contributes a predefined fraction proportion of the total output energy, thereby improving spatial uniformity. The framework efficiently identified four sets of grating parameters satisfying the above criteria. These are summarized in Table S1, with structural variations tailored to each subregion. Figure 2b shows the simulated intensity profiles from each out-coupler segment, demonstrating a spatially uniform output with 92% average uniformity, as calculated from pixel intensity statistics. These results confirm the effectiveness of slanted TiO_2 gratings and PSO-based optimization framework in AR waveguide design. They also highlight the importance of precise and scalable fabrication processes for practical implementations. To fulfill all process requirements, we present a comprehensive fabrication flow. Figure 3a illustrates the complete process for wafer-scale fabrication of TiO_2 slanted gratings, including IL, ICP, NIL and slanted etching via RIBE. Moreover, the process can be extended and optimized for DOW fabrication, as demonstrated in Figure S5.

All fabrication stages are critical to ensure high pattern accuracy, particularly the preparation of the Si master for NIL. IL was performed using a 266 nm DUV laser, and Htk-04B photoresist was selected for its high sensitivity and resolution in the DUV range. To suppress standing wave effects and improve fringe contrast, a 150 nm antireflection coating (AZkrfG7-200) was applied beneath the resist, as determined by optical interference modeling. The grating period was designed as 550 nm (see Figure 1c), with the IL beam incidence angle set to 14° according to the optical interference equation $P = \frac{\lambda}{2\sin(\theta)}$. An exposure dose of 17.86 mJ/cm^2 was selected to achieve a high contrast grating as shown in Figure 3b. The optical path setup schematic and AFM characterization of alternative doses are provided in Figure S6, demonstrating the dose-dependent response and validating the optimal dose selection. This precisely controlled exposure process establishes the foundation for subsequent pattern transfer.

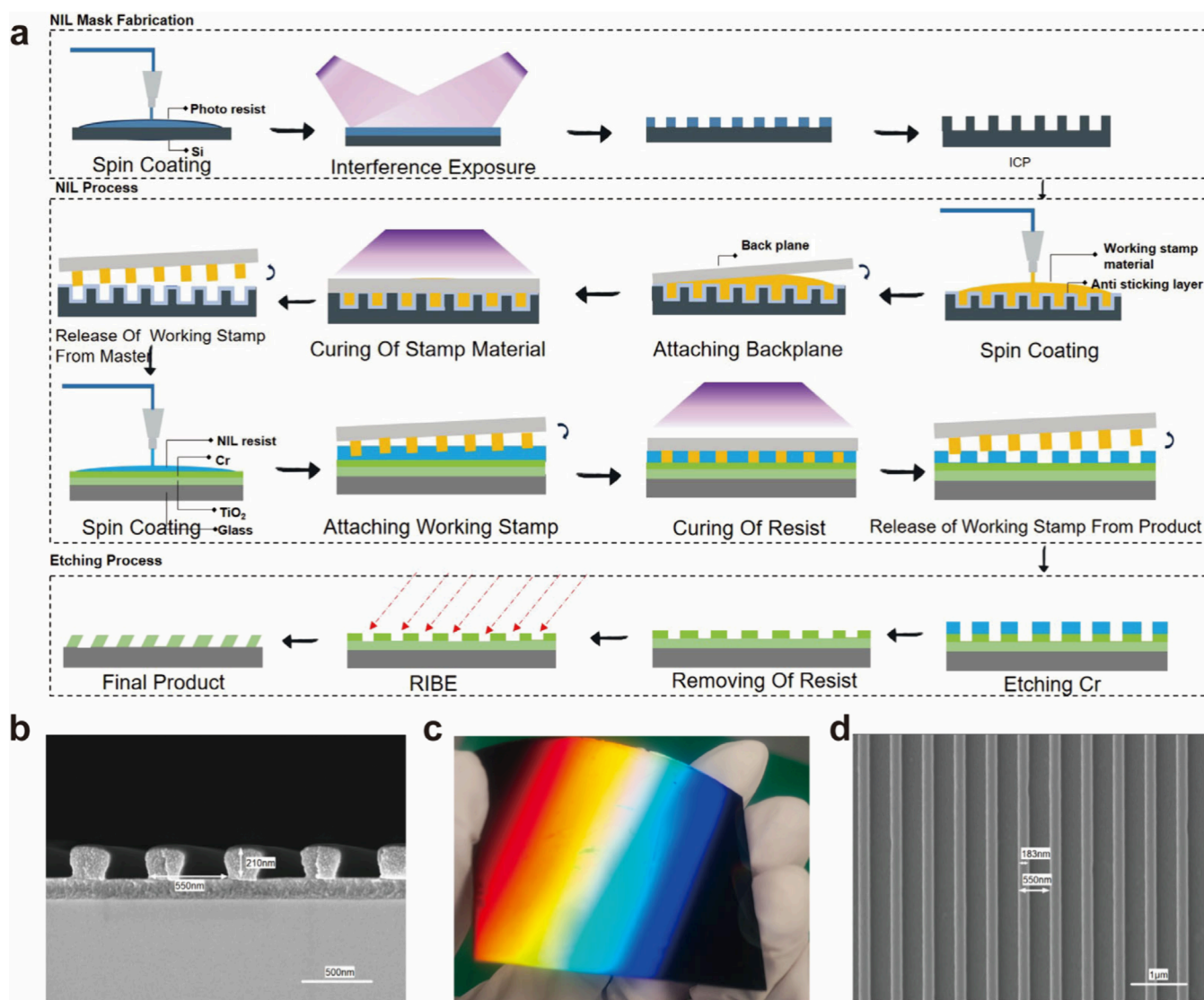


Figure 3. (a) Schematic of the fabrication process for scalable slanted TiO_2 grating. The process includes three major stages: I. fabrication of a Si master mold using interference lithography and ICP etching; II. pattern transfer onto a Cr-coated TiO_2 wafer via NIL; and III. final pattern transfer through a sequence of ICP Cr etching, photoresist removal, angled RIBE, and Cr wet etching. (b) Side-view SEM image of the interference lithography transferred pattern. (c) The image of the Si master mold after ICP etching and photoresist removal. (d) Top view SEM images of the NIL-transferred pattern.

As demonstrated in Figure S3a,b, precise control over the fill factor is critical for ensuring the grating performance. Due to postexposure fill factor deviation, a two-stage ICP etching process was implemented to modulate the grating ridge width. To address postexposure deviations in the photoresist (PR) profile, we first implemented a trimming etch to laterally narrow the PR ridge while maintaining its vertical dimension. This step enabled high-precision control of the grating line width, as illustrated in Figure S7a. Subsequently, we conducted the main pattern transfer etch, in which the refined PR structure was used as a mask to etch into the Si substrate. This anisotropic etching step achieved high selectivity and vertical sidewall fidelity, resulting in deep silicon trenches with straight profiles, as shown in Figure S7b. Finally, the residual PR was removed by O_2 plasma ashing, yielding the Si master shown in Figure 3c.

For the NIL process, the Si master was first treated with an antistick layer to prevent damage during demolding and to enable reliable pattern transfer. A flexible stamp was then

fabricated and used to transfer the grating pattern onto a Cr-coated TiO_2 substrate via thermal-assisted NIL with SUN-1611N-2 resist. To ensure high-fidelity pattern transfer and minimize residual layers, the resist thickness was controlled to 140 nm based on the constant-volume imprinting assumptions. Simultaneously, thermal softening at 45 °C improved the flowability of the resist, ensuring uniform filling of the recessed features. Before imprinting, the Cr-coated wafer was treated with O_2 plasma to enhance interfacial adhesion and suppress resist delamination. Figure 3d presents top-view SEM images of the NIL result, confirming high pattern fidelity. Images of the working stamp and wafer are shown in Figure S8a,b. To evaluate process reproducibility, the NIL process was repeated on 20 2-in. wafers and AFM measurements confirmed uniform pattern fidelity across 20 samples (Figure S9), validating the robustness and scalability of the process.

After NIL replication was completed, the Cr hard mask layer was patterned using ICP etching and the PR was also removed through ashing (Figure S7c). Following these steps, RIBE was

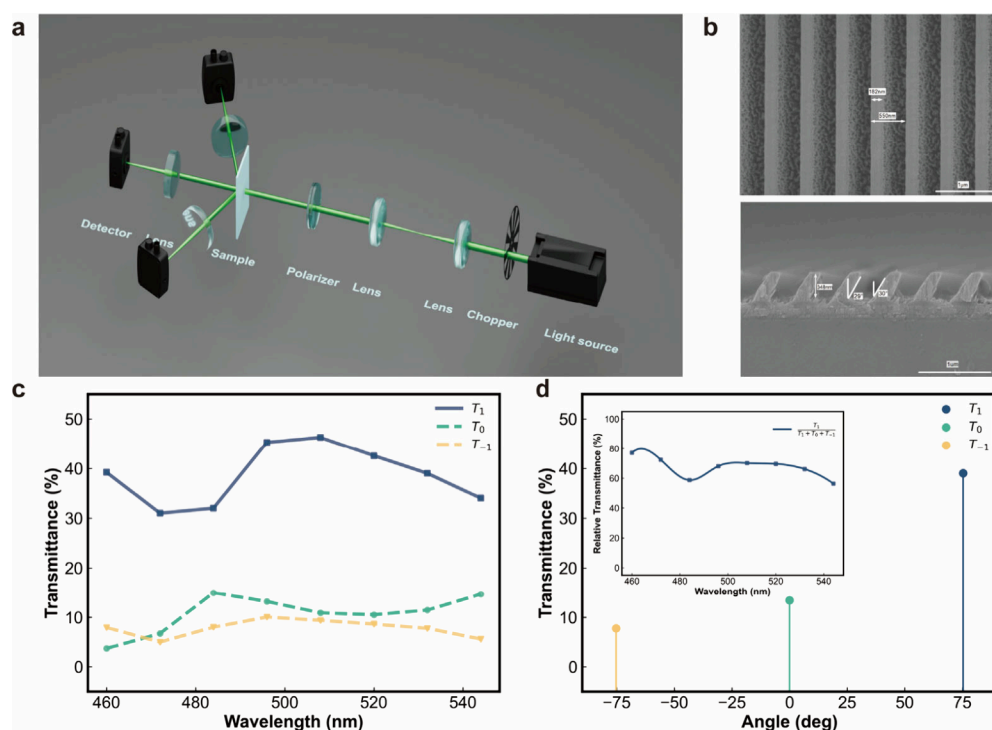


Figure 4. (a) Schematic of the optical measurement setup for diffraction efficiency characterization, including collimated laser source, rotational sample stage, and photodetector. (b) Top- and side-view SEM images of the slanted TiO₂ gratings fabricated via RIBE, showing a consistent sidewall angle of 30° and fill factor close to 0.32. (c) Wavelength-dependent transmittance spectra for the T₁, T₀, and T₋₁. (d) The transmittance of different orders at different emission angles and the corresponding relative transmittance of the + first diffraction order.

employed as the final and most critical process to fabricate slanted TiO₂ sidewalls. Unlike conventional directional ICP etching techniques which require large-angle tilted holders that preclude automation, the RIBE system is ideally suited for asymmetric structures (Figure S10). As schematically illustrated in Figure S10a, the ion beam is aligned along the *x*-axis while the wafer rotates about the *z*-axis, enabling precise control of the ion incidence angle (θ). SF₆ reactive gas was employed to generate chemically reactive species, including SF₅⁺ ions and fluorine radicals (F). The SF₅⁺ ions physically sputter the TiO₂ surface and disrupt Ti–O bonds, while the F radicals chemically enhance the etch rate through the formation of titanium fluoride compounds. This dual mechanism, combining physical sputtering and chemical etching enables efficient and controllable pattern transfer into the TiO₂ layer.^{34,35} More details are provided in the etching mechanisms section of the Supporting Information. As shown in Figure 4b, the resulting grating structure demonstrates a well-defined 30° sidewall angle, closely matching the design specification. To further assess the process robustness for DOW applications, slanted TiO₂ gratings with varying angles (0° to 45°) were fabricated using RIBE, as detailed in Figure S11.

Under illumination at 532 nm (Figure 4a), a strong first-order diffracted spot was observed, consistent with the design objective. The measured signal current for this spot reached 161 nA, corresponding to 39.1% of the incident intensity (412 nA). As shown in Figure 4c, T₁ remained dominant across the spectral range, achieving a relative intensity ratio of $\sim 66\%$ ($\eta = \frac{T_1}{T_1 + T_{-1} + T_0}$) at 532 nm. Figure 4d further confirms the engineered directional transmission, showing the angular distribution of diffraction efficiency up to 75°. The measured

angular profile exhibits a sharp peak around T₁, indicating strong angular selectivity and effective suppression of the T₀ and T₋₁. This angular response shows strong agreement with simulation and further validates the grating's ability to redirect incident light efficiently into large-angle diffraction channels, which is crucial for AR waveguide applications. In contrast to the RCWA simulation results (Figure 1c), which predicted T₁ of 80%, the experimental absolute efficiency (39.1%) is lower. However, the measured data exhibits the same key trend predicted by simulation: dominance of T₁ and suppression of T₀ and T₋₁ as Figure 4c shows. The reduced transmission efficiency is likely influenced by several factors not fully captured in the idealized simulation. We believe the main limitation arises from the deposition technique. We chose magnetron sputtering because of its compatibility with industrial applications, enabling direct scalability to larger wafer sizes. However, sputtered TiO₂ films may contain nanocrystalline domains rather than being fully amorphous.⁴¹ During subsequent fabrication steps, these nanocrystals can undergo local deformation or stress-induced rearrangements, thereby altering the optical properties of the film. Despite the reduced efficiency, our results experimentally validate the directional diffraction behavior targeted by our design. Moreover, a recent work has demonstrated the feasibility of TiO₂ slanted grating, which uses electron-beam evaporation to deposit TiO₂ and EBL to define more complex pattern, achieving up to 83.6% T₁.²⁰ However, such processes are inherently expensive, slow, and incompatible with wafer-scale production. This suggests that comparable results can be achieved with alternative deposition techniques. By contrast, our fabrication strategy, which integrates IL, NIL, and RIBE, offers a scalable, low-cost alternative while retaining the desired directional control. These results collectively highlight the

potential of our fabrication strategy for future AR waveguide research and applications requiring industrial scalability.

In summary, we have developed a scalable and robust wafer-level fabrication strategy for slanted TiO_2 gratings by integrating IL, ICP, NIL, and RIBE. This process enables precise control over grating geometry and slant angle across wafer-scale substrates. Furthermore, we investigated the underlying mechanisms of TiO_2 etching during RIBE, highlighting the interplay between physical sputtering and fluorine-enhanced chemical reactions. Experimental results demonstrate that T_1 dominates at a large diffraction angle of 75° under TM polarization, in good agreement with RCWA simulation results. We experimentally validated the fabrication of multiple grating geometries, laying the foundation for future integration into a DOW. These results highlight the potential of our platform for future angle-engineered diffractive components in AR waveguide couplers, holographic projection systems, and beam steering optics, enabling scalable production and practical deployment of high-performance diffractive optics in industrial AR applications.

In addition to these industrial applications, out-of-plane asymmetric nanostructures also enable novel physical phenomena, including bound states in the continuum (BIC) and chiral metasurfaces,^{42,43} which allow the manipulation of topological polarization singularities in momentum space. By providing a scalable wafer-level framework, our approach extends the accessible design dimension beyond conventional in-plane asymmetry^{44,45} and offers compatibility with other material systems such as Si_3N_4 and SiC. We believe this combination of practical manufacturability and expanded design freedom carries significant scientific value, opening new opportunities for both advanced photonic research and real-world optical technologies.

■ ASSOCIATED CONTENT

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.nanolett.5c04104>.

Experimental methods; RCWA and field simulations of slanted versus straight gratings (Figures S1–S3); waveguide configuration and grating parameter table (Figure S4, Table S1); fabrication strategy using IL, NIL, ICP, and RIBE with custom masks (Figure S5); IL setup and AFM resist profiles (Figure S6); SEM images of ICP trimming and etching (Figure S7); NIL stamps and transferred patterns (Figure S8); wafer-scale repeatability analysis (Figure S9); schematic illustrations of RIBE and ICP systems (Figure S10); TiO_2 etching mechanisms and SEM results (Figures S11–S12) (PDF)

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Author Contributions

The manuscript was written through contributions of all authors. All authors have given approval to the final version of the manuscript.

Notes

The authors declare no competing financial interest.

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